

Research Datacenter Security Plan

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Datacenters offer many benefits such as providing protection for sensitive information. They also help ensure that applications, programs and services are always up and available for use (Cirion). This demonstrates one of the Availability aspects of the CIA triad. Datacenters should also adhere and help uphold the other two elements, Confidentiality and Integrity. Throughout this report I shall go through the features that I have added.

I was inspired heavily by my workplace. I work as an intern at a hospital in their IT department. We have a server room onsite that helps us upkeep all our Virtual Desktop Infrastructures (VDI) and all related clinical applications needed to provide healthcare. We also help route calls from other hospitals and clinics associated with the Department of Health Services for the county of Los Angeles. Therefore, I am adopting some of the practices I observed while working there these past few months, from live camera feeds to cybersecurity training to logging practices.

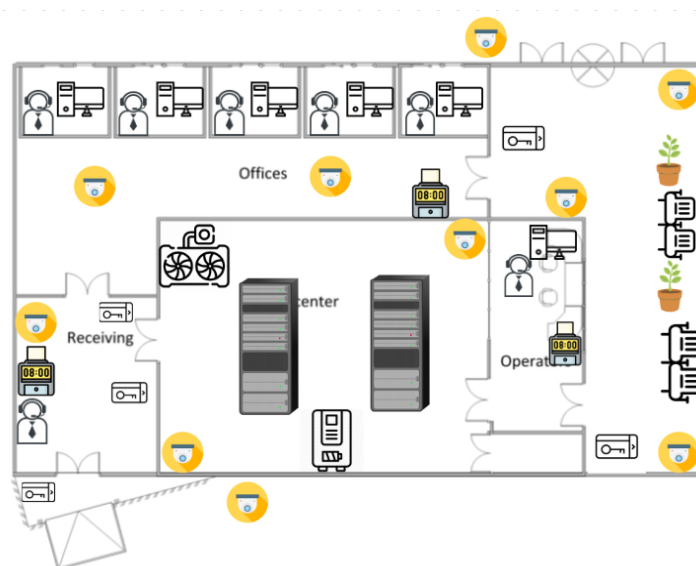
The entrance of the data center will have cameras posted outside. Every student or non-affiliated staff must check in with the operators first and state the reason for their visit. After which the Operators may buzz them in for any server room related purposes. The entrance area will have seating available in case employees may want to take their break there. The entrance area will also be used to receive guests in case there is a line of people queuing to enter the server room. Which brings us to our next point, there will be a limit on the number of people allowed to enter the datacenter. Accepting large groups into the server room makes it harder to monitor what is going on inside.

The inside of the server room will also have at least two cameras that will continuously broadcast inside the Operator room. Inside the datacenter room there will be server racks, computer room air conditioners (CRAC), routers, switches, firewalls, cables, modems, SSD's,

and UPS systems. All these components help facilitate flow and connectivity (Howell). We can delve into different companies such as CISCO, DELL or IBM for IT equipment. These companies offer the previously mentioned equipment such as switches, firewalls, routers, phones, servers, desktops, and laptops (Talented Technologies). Additionally, for cybersecurity related programs, the college may employ programs like Trellix. They may also adopt MFA practices every time an employee or student logs onto a workstation. Depending on the college they may either download Microsoft's authenticator app or Google's authenticator.

To enter the datacenter, the offices and receiving area will require a keycard to open the door. It will not be required to scan a keycard reader to exit these areas. My reasoning is that in the case of a fire, or some sort of electrical mishap, employees will not be trapped inside in the case of a keycard reader malfunction. However, there will be tablets to clock in and out for employees posted in the operator room, the offices, and the receiving room.

To gain access to the datacenter in the first place, students and employees must go through cybersecurity training. This training will be something that must be carried out yearly, or their access will be revoked. In the case that an unauthorized person tries to barge in, there will be an alarm system set in place that may alert campus police or local authorities.



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